

Lab 03 Template

- 1) **Take a screenshot of the five most common trigrams. Make sure you copy all the data including counts, distances, and common divisor.**

(10 total points, 2 points each trigram with full information)

```
Trigram: ELW
Count: 7
Positions: [203, 262, 478, 588, 658, 827, 943]
Differences: [59, 216, 110, 70, 169, 116]
GCD: 1

Trigram: OAS
Count: 5
Positions: [5, 348, 628, 763, 998]
Differences: [343, 280, 135, 235]
GCD: 1

Trigram: KWF
Count: 5
Positions: [26, 456, 711, 791, 851]
Differences: [430, 255, 80, 60]
GCD: 5

Trigram: JSM
Count: 5
Positions: [108, 373, 438, 568, 653]
Differences: [265, 65, 130, 85]
GCD: 5

Trigram: CGG
Count: 4
Positions: [164, 249, 814, 909]
Differences: [85, 565, 95]
GCD: 5

What is the likely key length? █
```

- 2) **Why do we analyze trigrams (three-character sequences) when attempting to break a Vigenère cipher? What information do the differences between their positions provide?**

(10 total points, 5 points each question)

Ans) Over here, we are analyzing trigrams to break a Vigenere Cipher because the same 3-letter word appears more than once, which means that the same part of the plaintext was encrypted using the same part of the key and since we know that the Vigenere key repeats, the repeated trigrams can basically help us in detecting patterns about how the key repeats itself.

The differences between the positions of the trigram actually tells about the distance between the 2 trigrams from each other in the ciphertext that has been given to us

- 3) **How does calculating the greatest common divisor (GCD) of position differences help in determining the key length?**

(5 total points)

Ans) Calculating the GCD helps us in determining the length of the key as the distances between the repeated trigrams are usually multiples of the keyword's length, hence the GCD basically finds the largest number that could divide all the distances.

- 4) **Based on your findings, what do you suspect the key length is? Justify your answer.**

(10 total points, 5 points length value, 5 points justification)

Ans) The length of the keyword is 5, since 3 out of 5 trigrams said that their GCD value is 5 based on their position differences, as the repeated trigram distances are usually the multiples of the keyword's length and since it appears as the GCD for most of the trigrams, it creates a suspicion that 5 might be the length of the key.

- 5) **Why do you assume the most frequent character is E? How does it help us with multiple shift-by-n ciphers?**

(10 total points, 5 points each question)

Ans) We assume that the most frequent character is E, since it is the most common alphabet used in the English language, so whenever a Caesar cipher encrypts the English text, it is assumed that the ciphertext that appears the most is basically the encrypted form of E.

It basically helps us with multiple shifts by n-ciphers, as once we split the ciphertext into different columns, each column acts like its own Caesar cipher, so if we assume that the

most frequent letter in each column basically represents E, then we can estimate the shift amount for that column, which later helps us in figuring out the keyword.

- 6) **Take a screenshot for *each* shift-by-n cipher**
(10 total points)

```
Analyzing Cipher 0:
Frequency for Cipher 0:
W: 22
G: 20
L: 19
F: 16
A: 14
O: 13
V: 13
J: 13
S: 12
M: 10
Z: 10
D: 9
Q: 6
K: 6
Y: 5
E: 4
X: 4
T: 3
N: 3
P: 2
H: 1
U: 1
B: 1
Corresponding alphabets:
W: Shifted Alphabet (setting E to W): STUVWXYZABCDEFGHIJKLMNOPQR
G: Shifted Alphabet (setting E to G): CDEFGHIJKLMNOPQRSTUVWXYZAB
L: Shifted Alphabet (setting E to L): HIJKLMNOPQRSTUVWXYZABCDEFGHI
A: Shifted Alphabet (setting E to A): STUVWXYZABCDEFGHIJKLMNOPQR
```

```
Analyzing Cipher 1:
Frequency for Cipher 1:
M: 20
Q: 20
A: 19
F: 19
U: 18
P: 13
Z: 13
D: 12
I: 9
K: 9
X: 9
T: 8
E: 8
S: 6
G: 6
R: 5
H: 3
J: 3
Y: 2
V: 1
W: 1
N: 1
O: 1
B: 1
Corresponding alphabets:
M: Shifted Alphabet (setting E to M): IJKLMNOPQRSTUVWXYZABCDEFGH
Q: Shifted Alphabet (setting E to Q): MNOPQRSTUVWXYZABCDEFGHIJKL
A: Shifted Alphabet (setting E to A): WXYZABCDEFGHIJKLMNQRSTUUV
```

```
Analyzing Cipher 2:
Frequency for Cipher 2:
W: 21
V: 20
M: 19
Q: 18
B: 17
I: 15
Z: 12
P: 12
A: 11
L: 9
T: 9
E: 9
C: 6
G: 6
O: 5
U: 4
D: 3
S: 2
N: 2
F: 2
J: 2
H: 1
X: 1
R: 1

Corresponding alphabets:
W: Shifted Alphabet (setting E to W): STUVWXYZABCDEFGHIJKLMNOPQR
V: Shifted Alphabet (setting E to V): RSTUVWXYZABCDEFGHIJKLMNOPQ
M: Shifted Alphabet (setting E to M): IJKLMNOPQRSTUVWXYZABCDEFGH
```

```
Analyzing Cipher 3:
Frequency for Cipher 3:
P: 24
E: 20
Z: 17
D: 15
L: 14
T: 14
H: 13
S: 12
Y: 12
C: 11
O: 9
F: 8
W: 8
J: 7
R: 4
Q: 4
G: 3
V: 3
X: 2
M: 2
I: 2
N: 1
A: 1
Corresponding alphabets:
P: Shifted Alphabet (setting E to P): LMNOPQRSTUVWXYZABCDEFGHIJK
E: Shifted Alphabet (setting E to E): ABCDEFGHIJKLMNOPQRSTUVWXYZ
Z: Shifted Alphabet (setting E to Z): VWXYZABCDEFGHIJKLMNQRSTU
A: Shifted Alphabet (setting E to A): CDEFGHIJKLMNOPQRSTUVWXYZA
```

```
Analyzing Cipher 4:
Frequency for Cipher 4:
E: 22
S: 22
R: 20
X: 19
I: 17
M: 15
L: 14
A: 12
H: 10
Q: 7
P: 7
W: 7
Y: 6
V: 6
C: 5
F: 4
K: 3
J: 3
T: 2
B: 2
O: 1
Z: 1
N: 1
Corresponding alphabets:
E: Shifted Alphabet (setting E to E): ABCDEFGHIJKLMNOPQRSTUVWXYZ
S: Shifted Alphabet (setting E to S): OPQRSTUVWXYZABCDEFGHIJKLMN
R: Shifted Alphabet (setting E to R): NOPQRSTUVWXYZABCDEFGHIJKLM
```

- 7) **Decrypt the ciphertext using the potential key values you found in question 5. Show all iterations, the final keyword, and the final plaintext.**
(30 total points, 20 points for all iterations, 5 points for correct keyword, 5 points for correct plaintext)

Guess 1

```
Enter the keyword you think will decrypt the ciphertext: SISLA

Decrypted Plaintext:
INKSXWAEYPJHOQAHHEEMAXEVECEUENHYHEDXESEYKEOHBCUARDMTORTODOAOWLQTMTEBLQEEDSFUXIIRCIYSYRZYVIDMHEELMPEHWLURIVIHYSUKETLAXIWL
EVUIPLJELPOADOFHOHOSTRSCIEHJQOVHOASSYMELSLCOYUVIRCDIKHXBIOEMJSXHIBAWTRYGLTPYKIIITILEPQSNMWHXIJJHIWSHLHWEIERDMGMDAQNR
AFUNIXXJOCOYFYXHFIAVTCMAWOZURENHEUVTMCESNIQRXHAQSXHVEUKHMTWENRQSHLOOYJYITJQVQWLIPUARDHYEAIIXAWMMBEMFXXEAOVBDAWUNHIRWIH
WEDNEBIDEBTXEYSUPESXLSITMNXEAOVTSXHEJWISGHEEMMTORTILERWEDNEDSJHSEDYQOVUCEUWUYSUEBRIAHOKROAMHETCEUQEEDTSMIQNHQYHLSVITL
ESDLCWEHWSRXFXMGLJIRGJERAHIHEZEVOOYGSJHETWMHIRIYLPFSBLSWREBSDCIPVOQYSIDXEMSRVEWWMCAPOZUYSUILEVYRYGLTPYKIIITILEPQSNMWHX
LMAEMTWJHILEITRIKXTMFXXEAOVBDAWUNHIRWIHWEDNEBIDEBTXEYSUMVLTETQRXYAQSSVIHARDSKRXIQUOREHTLWEITLRSKGLIHMARNEXPDCUUNWJFS
REMHMLIQNHDMUWMTLQSQIPUIJTJLWUSRPTWESIDDMNKYDAARDAFERUXXTSOORYMWHXNINTXOCEUREBJTSYSKRMGLJNIXXJOCOYFYXHIMOVLMHAWERTIRGMTWE
NRQBINTXOCEUMFXXETAJYAAWEVIREDDSUJJIQESDEERXWESXXRSUKXIHWEDNEHSBDCOYZUWTJEREWLYLARTDMEAYTLAWCIPEMVTLEAERPDAQSINHYNK
IHMARNEREREBJTSYSKIJTLWUSRPTWESIDDMNKYDAARDAFERUXXTSOORYMARNEREREBJTSYSK
```

This was basically my first guess basically after determining the length of the keyword over here. In this case, I assumed the most frequent ciphertext letter in each column represented the English letter 'E', which yielded the initial keyword 'SISLA', but the output was mostly random, indicating that some key letters were incorrect.

Guess 2

```
Enter the keyword you think will decrypt the ciphertext: SMSLA

Decrypted Plaintext:
IJKSXWAEYPPFHOQADHEEMXVEVEYUENDYHEDTESEYGEOHBYUARDIORTKDOAWHQTMTABLQEADSFUTIIIRCEYSYRVVYVIDIHEELIPEHWHURIVEHYSUGETLATIWL
ERUIPLFELPOWDOFODOSTROCIWEDJQORHOASOYMELOLECOUUVIRYDIKHTBIOEIJXHEBAWNTYGLTYKIIITILELQSNXIWHXIFJHIWOHLHWAIERDIDGMWQNR
ABUNIXTJOCOYFYXHEFAVTYMAWOVURENDEUVTICESNEQRXHWQSHREUKHITWENRQSHLDOOYJUITJQVQWLILUARDHYEAITXAWMIBEMFTEAORBDASUNHINWIH
WADNEBEDEBTTEYSULESXLOITMNTXEAORTSXHAJWISCHEEMITORTELERWADNEDSJHMSADYQORUCEUSUYSUABRIADOKROWMHETYEUEADTSMEQNHOHLVSVEITL
EODLCWAHWSRTXFMGLJIRGFERAHEHEZEROOYGOJHETSMHIREYLPFOBSLWNEBSDYIPVOMYSIDTEMSRREWMOICAPOVUYSUELEVYNYGLTYKIIITILELQSNXIWHX
LIAEMTSJHILAITRIGXTMFTXEAORBDASUNHINWIHWADNEBEDEBTTEYSUIVLEPQRXYWQSSVEHARDOKRXIMUOREAHTLWAILROKGLIDMARNAXOPDYEUUNWJFS
RAMHMLEQNHDIUWMTQSQILUIJTJHUSRLTWESEDDMNGYDAANDAFENUXXTOOOYRIWHXNENTXOYEUREXJTSYOKRMGLJNIXTJOCOYFYXHEMOVLDMAWENTIRGITWE
NNQBINENTXOYEUMFTXETARJYAASEVIRADDSURJJIQODEERTXWESTXRSUGXIHWADNEHOBDCOUZUWTFEREWHYLIANTDMEWYTLASCIPEIVTLEWRPDWQSINDYNK
IDMARNAREREXJTSYOKIJTHUSRLTWESEDDMNGYDAANDAFENUXXTOOOYIDMARNAREREXJTSYOK
```

This was my next guess. Over here, I basically changed the ciphertext 1 from I to M since ciphertext letters M and Q had the same frequency rate, so that's why I made a change in that column and observed that the plaintext improved slightly, which indicated that M should be used instead of I in the ciphertext 1 column.

Guess 3

```
Enter the keyword you think will decrypt the ciphertext: SMRLA

Decrypted Plaintext:
IJLSXWOBEPFIOQADIEEMWYEVEYFUENDZHEDTFSEYGFHOBVYVARDIUORTKEOAWHRTMTACLQEAESFUTJIRCEZSYRVZVIDIIEELIQEHWHVRIEYISUGFTLATJWL
ERVIPFLFPOWEOFODPSTRODIWEDKQOQRIOASOZMELOMECOUVVIRYIEIKHTCIOEIKSXHECAWTNZGLTLZKIIITJTLRLSXNIXHXIFKHIWOILHWAJERDIEGMWNRN
ABVNIXTKOCOUZFXHEGAVTYNAWOVVRENDUFVTIDESNERRXHWRSXHRFUKHIUWENNRHSLDPOYJUJTJQVQWLILVARDDEZAITYAWMICEMFTYEAORCDAASVNHINXIH
WAENEBEEEBTTFYSULFSXLOJTMNTYEAORUSXHAKWISCIEEMIUORTEMERWAENEDOKHMSAEYQORVCEUSVYSUACRIADPKROWNHETYFUQEAETSMERNHOUILSVEJTL
EOELCWAISWRTYFMGKIRGFFRAHEIEZERPOYGOKHETSNHIREZLPFOCLSWNFSBDDYJPVOMZSIDTFMSRRFWOIDAPOVVYSUEMEVYNZGLTLZKIIITJTLRLSXNIXHX
LIBEMTSKHLAJTRIGYTMFTYEAORCDAASVNHINXIHWAENEBEEEBTTFYSUIWTLTPRRXYWRSSVEIARDOLRXIMVOREAITLWAJTLROLGLIDNARNAYOPDYFUNUSKFS
RANHMLERNHDIVWMTQRSQILVIJTHVWSRLUWESEEDMNGZDAANEAFENVXTOPOYRIXHXNEOTXOYFUREXKTSVOLRMGHKNIXTKOCOUZFXHENOVLDNAWENUIRGIUWE
NNRBINEOTXOYFUMFTYETARKYAAAFVIRAEDSURKIQOEERTYWESTYRSUGYIHWAEHOCDCOUAUWTFREWHZLIANUDMEWZTLASDIPEIWTLEWFRPDWRSINDZNK
IDNARNASEREXKTSYOLIJTHVWSRLUWESEEDMNGZDAANEAFENVXTOPOYIDNARNASEREXKTSYOL
```


Next, I moved to ciphertext 2 column, where I tried mapping the next most frequent letter over there to E, which in turn changed the key letter from S to R, which produced SMRLA, which slightly improved the readability of the plaintext, however, it was not fully readable yet.

Guess 4

```
Enter the keyword you think will decrypt the ciphertext: SMILA
I
Decrypted Plaintext:
IJUSXWOKEYPFROQADREEMWHEVEYOUENDIHEDTOSEYGOOHBYEARDIDORTKNOAWHATMTALLQEANSFUTSIRCEISYRVIVIDIREELIZEHWHERIVERYUGOTLATSWL
EREIPLFOLPOWNOFODYSTROMIWEETOQORROASOIMELOVECOUEVIRYNIKHTLIEITXHELAWTNIGLTLIKIITSTLELASXNIGHXIFTHIWORLHWASERDINGMDWANR
ABENIXTTOCOUIFXHEPAVTYWAWOVERENDOUVTIMESNEARXHWASXHROUKHIDWENNAHSLDYOVJUSTJORAWLLEARDDEIAITHAWMILEMFTHEAORLDAASENHINGIH
WANNEBENEBTTOYSULOSXLOSTMNTHAORDSXHATWISCREEMIDORTEVERWANNEDOTHMSANYQORECEUSEYSUALRIADYKROWWHETYOUQEANTSMEANHOURLSVESTL
EONLCWARWSRTHFMGHTIRGFORAHEREZERYOYGOTHETSWHIREILPFOLLWNBSDSPVOMISIDTOMSRROWWOIMAPOVEYSUEVEVYNIGLTLIKIITSTLELASXNIGHX
LIKEMTSTHILASTRIGHTMFTHEAORLDAASENHINGIHWANNEBENEBTTOYSUIFTLEPARXYWASSVERARDOURXIMEOREARTLWASTLROUGLIDWARNAHOPDYOUNUSTFS
RAWHMLEANHDIEMTHASQILEIJTHEWSRLDWESENDMNGIDAANNAFENEXXTYOYRIGHXNEXTXOYUREXTTSYOURMGHTNIXTTOCOUIFXHEWOVLDAWAWENDIRGIDWE
NNABINEXTXOYUUMFTHETARTYAASOVIIRANDSURTIQEONEERTHWESTHRUGHIHWANNEHOLDCOUJUWTFORWHILIANDDMEWITLASHIPEIFTLEWOPDASINDINK
IDWARNABEREXTTSYOUIJTHEWSRLDWESENDMNGIDAANNAFENEXXTYOYIDWARNABEREXTTSYOU
```

Now, I tried to map the next frequent letter (M) in that same column, since the previous did not produce many english language words, which basically changed the third key from R to I, which gave me SMILA and when I tried to decrypt it, I observed that the output that I got was mostly readable, which gave me a hint that we might have to change only 1 letter to figure out the complete plaintext.

Guess 5

```
Enter the keyword you think will decrypt the ciphertext: SMILE
Decrypted Plaintext:
IJUSTWOKEUPFROMADREAMWHEREYOUANDIHADTOSAYGOODBYEANDIDONTKNOWWHATITALLMEANSBUTSINCEISURVIVEDIREALIZEDWHEREVERYOUGOTHATSWH
EREILLFOLLOWNOBODYSPROMISED TOMORROWSOIMALOVEYOU EVERYNIGHTLIKEITSTHELASTNIGHTLIKEITSTHELASTNIGHTIFTHEWORLDWASENDINGIDWANN
ABENEXTTOYOUIFTHEPARTYWASOVERANDOURTIMEONEARTHWASTHROUGHIDWANNAHOLDYOUJUSTFORAWHILEANDDIEWITHASMILEIFTHEWORLDWASENDINGID
WANNABENEXTTOYOULOSTLOSTINTHEWORDSTHATWESCREAMIDONTEVENWANNADO THIS ANYMORE CAUSEYOUALREADYKNOWWHATYOU MEANTOMEANDOURLOVESTH
EONLYWARWORTHFIGHTINGFORWHEREVERYOUGOTHATSWHEREILLFOLLOWNOBODYSPROMISED TOMORROWSOIMALOVEYOU EVERYNIGHTLIKEITSTHELASTNIGHT
LIKEITSTHELASTNIGHTIFTHEWORLDWASENDINGIDWANNABENEXTTOYOUIFTHEPARTYWASOVERANDOURTIMEONEARTHWASTHROUGHIDWANNAHOLDYOUJUSTFO
RAWHILEANDDIEWITHASMILEIFTHEWORLDWASENDINGIDWANNABENEXTTOYOURIGHTNEXTTOYOU NEXTTOYOURIGHTNEXTTOYOUIFTHEWORLDWASENDINGIDWA
NNABENEXTTOYOUIFTHEPARTYWASOVERANDOURTIMEONEARTHWASTHROUGHIDWANNAHOLDYOUJUSTFORAWHILEANDDIEWITHASMILEIFTHEWORLDWASENDING
IDWANNABENEXTTOYOUIFTHEWORLDWASENDINGIDWANNABENEXTTOYOUIDWANNABENEXTTOYOU
Do you want to try another keyword? (yes/no):
```

Next, I tried to change to the last ciphertext column, where I changed the key from A to E since 'SMILE' is a real English word, unlike the words that were produced from the other frequencies in this group (like SMILA, SMILO, and SMILN). Later on, when I tried to decrypt the key using the keyword SMILE, I observed that I got a plaintext that was completely readable. Hence, the correct keyword to decrypt this ciphertext is SMILE.

- 8) **What strategies can you use to refine your keyword guesses if the plaintext is not immediately recognizable?**

(5 total points)

Ans) I try refining the keyword by just changing one key letter at a time instead of changing the whole keyword. I just tried mapping the second or the third most frequent ciphertext letters to E instead of just trying the most frequent one every time, in some columns, then printed out the ciphertext to check whether I was able to see at least some common English language words in the plaintext.

- 9) **Submit your documented python code as <netid>lab03_part01.py to canvas.**

(10 total points for working code. No points awarded if the code is copy/pasted from someone else.)